

EXP 27 :

Aim:

To write a Python program to display a bar chart showing the popularity of different programming languages.

Algorithm:

1. Import the matplotlib.pyplot library.
2. Store the programming languages and their popularity values in lists.
3. Use plt.bar() to create a vertical bar chart.
4. Add title, X-axis label and Y-axis label.
5. Display the chart using plt.show().

Code:

```
import matplotlib.pyplot as plt

languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']

popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]

plt.bar(languages, popularity)

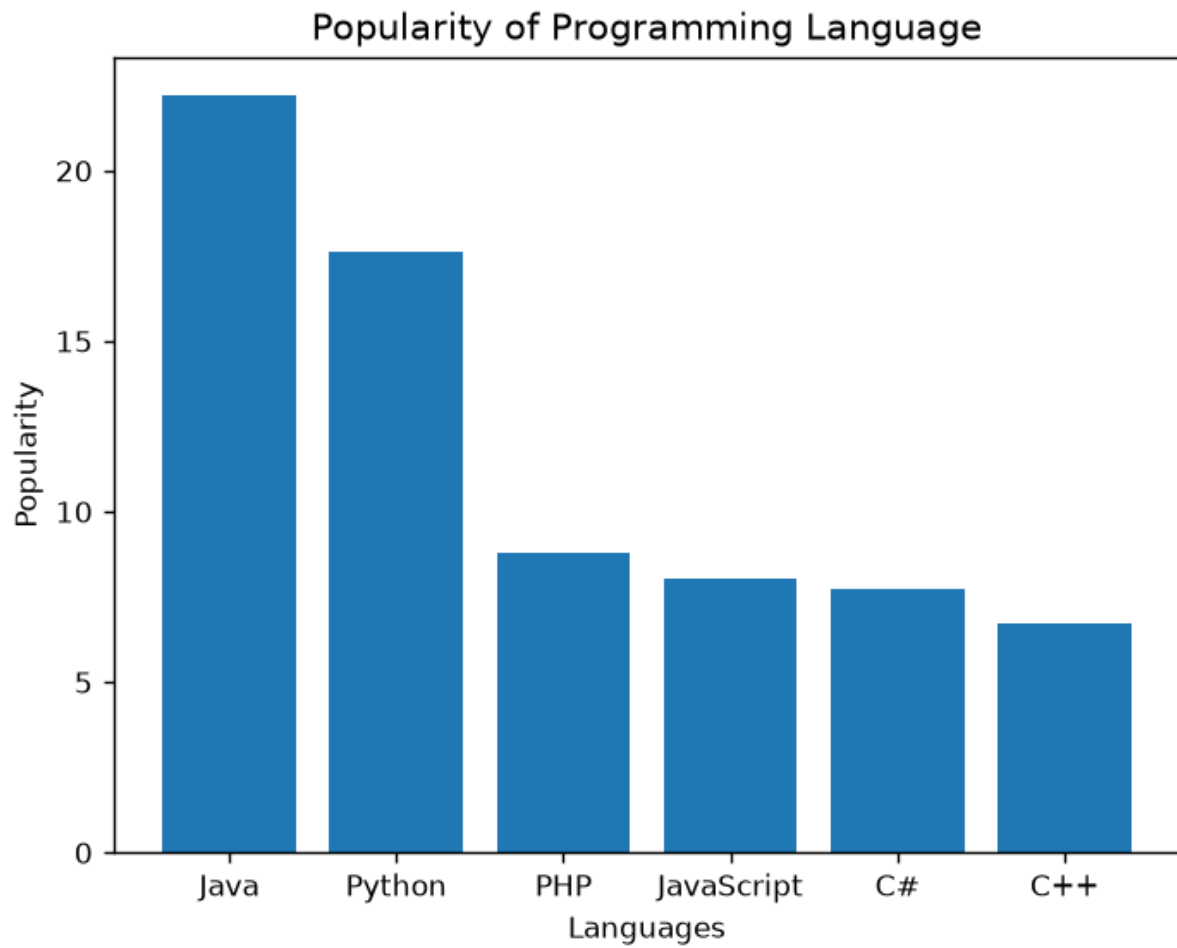
plt.title('Popularity of Programming Language')

plt.xlabel('Languages')

plt.ylabel('Popularity')

plt.show()
```

Output:



Result:

Thus, the Python program was successfully executed and the vertical bar chart displaying the popularity of programming languages was obtained.

EXP 28 :

Aim:

To write a Python program to display a horizontal bar chart showing the popularity of different programming languages.

Algorithm:

1. Import matplotlib.pyplot.
2. Store the languages and popularity values in lists.
3. Use plt.barh() to create a horizontal bar chart.
4. Add title and axis labels.
5. Display the chart using plt.show().

Code:

```
import matplotlib.pyplot as plt

languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']

popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]

plt.barh(languages, popularity)

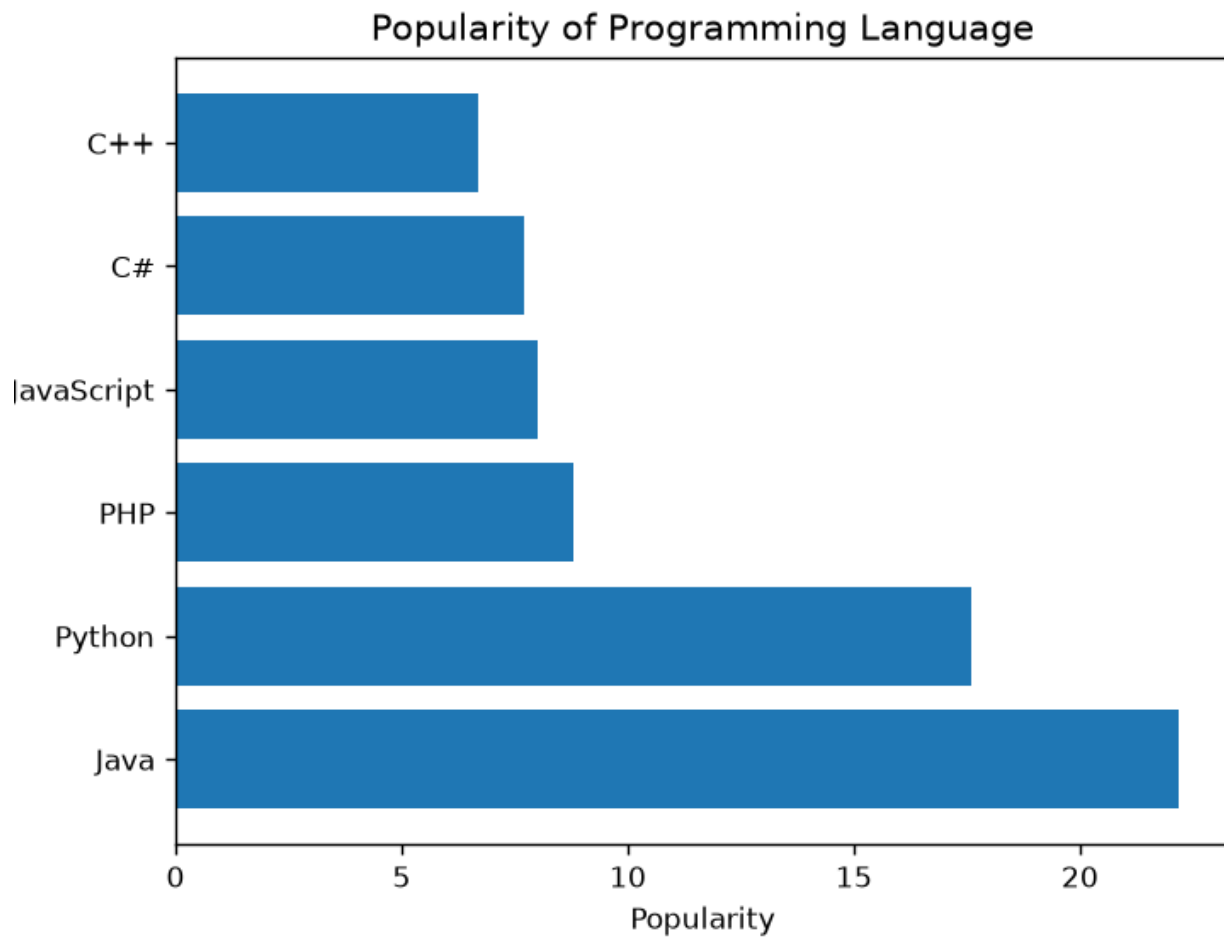
plt.title('Popularity of Programming Language')

plt.xlabel('Popularity')

plt.ylabel('Languages')

plt.show()
```

Output:



Result:

Thus, the Python program was successfully executed and the horizontal bar chart displaying the popularity of programming languages was obtained.

EXP 29 :

Aim:

To write a Python program to display a bar chart of programming language popularity using different colors for each bar.

Algorithm:

1. Import matplotlib.pyplot.
2. Store the languages and popularity values in lists.
3. Create a list containing different colors.
4. Use plt.bar() with the color parameter.
5. Add title and axis labels.
6. Display the chart.

Code:

```
import matplotlib.pyplot as plt

languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']

popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]

colors = ['red', 'black', 'green', 'blue', 'yellow', 'cyan']

plt.bar(languages, popularity, color=colors)

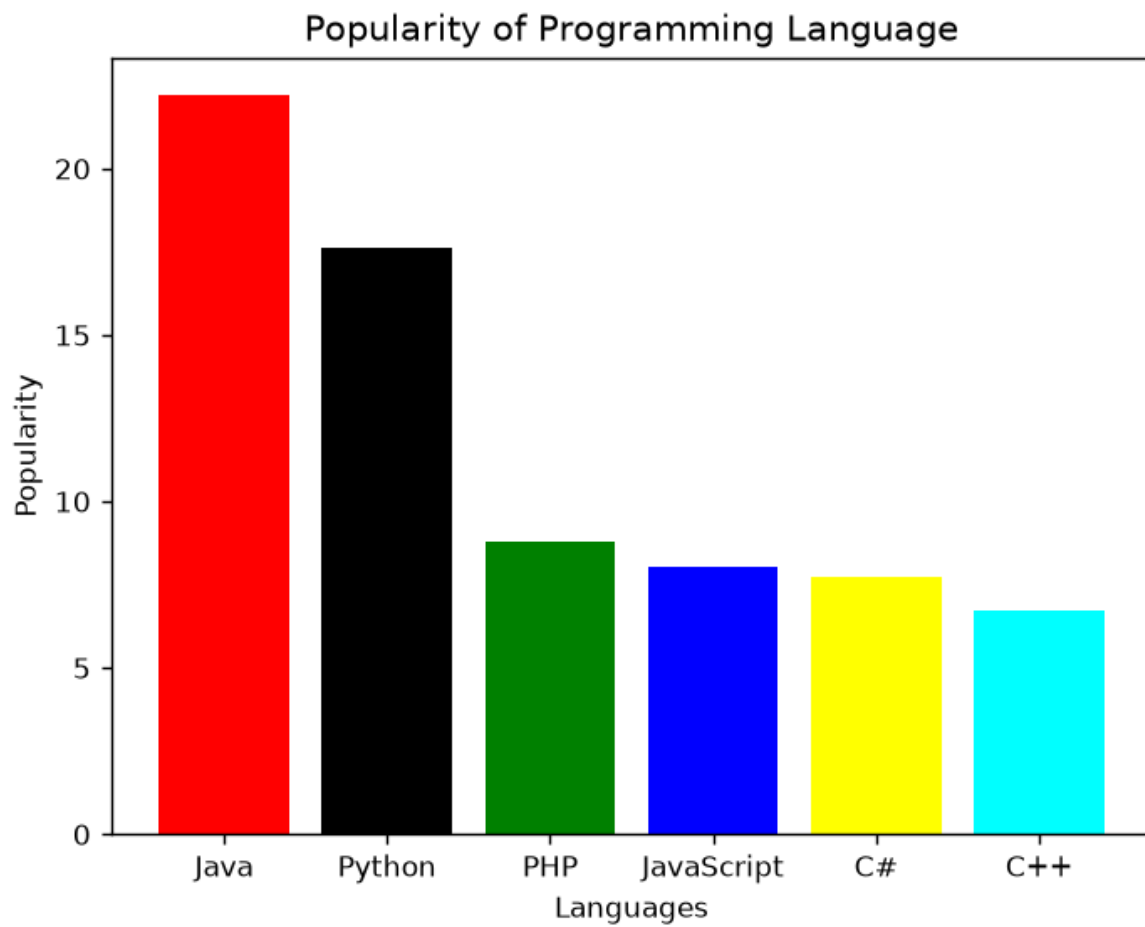
plt.title('Popularity of Programming Language')

plt.xlabel('Languages')

plt.ylabel('Popularity')

plt.show()
```

Output:



Result:

Thus, the Python program was successfully executed and the bar chart with different colors for each programming language was obtained.